Access-Constrained Clarification for Efficient and Safe Service in Text-to-SQL (ACCESS) Supplementary Analyses

This supplement accompanies ACCESS and uses the same request populations, pipeline names, and order-aware metrics as the main paper. It reports analyses of the frozen 2,800-request evaluation cohort; Section S16 documents its construction and planned-population completion sensitivity. Sections S1–S9 concern the four main pipelines; Sections S10–S12 add three completed module variants on the same requests. Section S13 examines exact output columns; S14 reports outcome-conditioned effort and SQL-stage accounting; S15 reports paired BSS intervals and weighted-harmonic sensitivity. Section S17 details the controlled-specificity construction and its validation. BSS consistently denotes the harmonic mean of SafeEX and SHR. All analyses reuse recorded SQL without new model inference during rescoring. Scores are defined on $[0, 1]$; reported rates are percentages and score differences are percentage points; ranges denote unresolved-evidence bounds unless explicitly labeled as bootstrap intervals. The main reference partition has $|A| = 1,402$ benchmark-allowed requests, $|P| = 80$ *recoverable denied requests* with a nonempty legal-output reference, and $|Z| = 1,318$ benchmark-denied requests without such a reference. Here A, P, and Z are disjoint request-index sets, $|S|$ denotes set cardinality, $E = A \cup P$ is the recovery cohort, and $\mathcal{I}_{\text{den}} = P \cup Z$ is the handling cohort. As in the main paper, P is the reference-positive benchmark-denied cohort; *recoverable denied requests* is a short name for this same fixed set.

The supplementary analyses follow the main results. Sections S2–S4 and S15 expand the main comparison in Section V-A (Tables II–III); S10–S12 expand the ablations in Section V-B (Table IV); S5–S6 explain the outcomes in Section V-D (Table V); S9 and S14 support the interaction-cost analysis in Section V-E (Table VI); S8 and S13 support the reference and exact-output checks in Section V-F (Tables VIII and VII, respectively). Sections S16–S17 document the cohort and specificity construction in Section IV-A.

S1 Fixed Reference Construction and Validation

The original benchmark labels are treated as ground truth for the request-level evaluation strata and retained for all pipelines. The fixed output reference is constructed independently of candidate SQL and scores. Reference witnesses are checked for permissions, complete result tuples, duplicate multiplicity, and required ordering; candidate SQL is checked separately under the same role policy.

The construction examines all 2,234 benchmark-denied output units and identifies 80 reference-positive requests. Two human reviewers independently examined all 1,318 reference-negative requests; neither identified additional legal service preserving the original requirements. The declared reference family retains the required result tuples, dependencies, and final ordering.

Current-database witness checks establish the required output for all 80 references. Two human reviewers independently assessed these references, followed by a final consolidated assessment that accepted all 80 (100%). The structurally supported partial-output subset contains 47 instances. The diagnostic subsets in Section S8 distinguish retained output positions, operator structure, and output content.

Grouping-label outputs occur in `pets_1/258` and `singer/44`; 12 cases in four clusters require changed Gold ordering and belong to Z. Sections S9 and S8 report human confirmation and operator/output diagnostics, respectively.

S2 Safe Handling Combines Refusal and Authorized Delivery

The four-way refinement interface follows Section II of the main paper: full and partial handoffs proceed to SQL generation, refusal ends service, and unresolved records a refinement that has not reached a justified terminal decision. Evaluation uses the recorded outcome of each request. As in the main paper, ref_i and safe_i are binary indicators of terminal refusal and delivery of a valid permissions-cleared SQL candidate, respectively. These outcomes are mutually exclusive; safe delivery does not itself require successful result execution or matching. On P, a recorded whole-request refusal receives SHR credit but zero recovery credit; an authorized SQL delivery receives recovery credit only when the execution-result checks pass. The component rates are

$$\text{CRR}_Z = \frac{\sum_{i \in Z} \text{ref}_i}{|Z|}, \qquad \text{SDR}_P = \frac{\sum_{i \in P} \text{safe}_i}{|P|}, \qquad \text{SHR}_P = \frac{\sum_{i \in P} (\text{ref}_i + \text{safe}_i)}{|P|}.$$

Safe delivery rate excludes refusals, whereas SHR_P includes them. For any nonempty evaluation subset $\Omega \subseteq E$, Full_Ω is the fraction with request-level column-recovery score $c_i = 1$: every target column must match as tuples with complete rows, duplicates, final order, and whole-query authorization. Full is complete output-reference recovery; extra authorized candidate columns remain unpenalized. Equal empty results alone retain conservative $[0, 1]$ bounds on column correspondence. If only one result is empty, the row mismatch scores zero.

Table S1: Safe-handling decomposition

Model	Pipeline	CRR_Z	Ref_P	SDR_P	SHR_P	Full_P
DeepSeek Flash	Direct	63.58	38.75	40.00	78.75	7.50
DeepSeek Flash	Serial	58.35	55.00	26.25	81.25	5.00
DeepSeek Flash	Policy	28.45	18.75	36.25	55.00	16.25
DeepSeek Flash	ACCESS	88.01	18.75	73.75	92.50	48.75
Qwen2.5-14B	Direct	24.81	20.00	33.75	53.75	18.75
Qwen2.5-14B	Serial	28.15	30.00	12.50	42.50	2.50
Qwen2.5-14B	Policy	28.53	18.75	22.50	41.25	8.75
Qwen2.5-14B	ACCESS	88.01	18.75	61.25	80.00	40.00
Arctic R1-7B	Direct	0.08	0.00	23.75	23.75	3.75
Arctic R1-7B	Serial	0.46	3.75	8.75	12.50	0.00
Arctic R1-7B	Policy	28.45	18.75	15.00	33.75	3.75
Arctic R1-7B	ACCESS	88.01	18.75	51.25	70.00	33.75
Qwen2.5-7B	Direct	38.85	22.50	20.00	42.50	7.50
Qwen2.5-7B	Serial	51.59	43.75	8.75	52.50	1.25
Qwen2.5-7B	Policy	28.45	18.75	10.00	28.75	2.50
Qwen2.5-7B	ACCESS	88.01	18.75	73.75	92.50	46.25
Qwen3.8-27B	Direct	77.39	58.75	25.00	83.75	3.75
Qwen3.8-27B	Serial	62.97	53.75	25.00	78.75	3.75
Qwen3.8-27B	Policy	28.45	18.75	25.00	43.75	10.00
Qwen3.8-27B	ACCESS	88.01	18.75	73.75	92.50	48.75

Note. **ACCESS combines correct refusal with more safe deliveries on recoverable denied requests.** Five-model denied-handling decomposition. Ref_P is the refusal rate on P. $\text{Ref}_P + \text{SDR}_P = \text{SHR}_P$; Full_P additionally requires complete correct results.

Table S1 shows ACCESS's highest correct-refusal rate on Z and safe-delivery rate on recoverable denied requests (P) within every model. Its frozen frontend supplies 1,160 refusals on Z and 15 on P across models; safe SQL counts on P are 59, 49, 41, 59, and 59, while complete recoveries number 39, 32, 27, 37, and 39. Thus the workflow combines rejection of unserviceable requests with authorized partial deliveries, but useful output still depends on SQL realization. The gap between safe and complete deliveries measures that downstream distinction under the same frontend decisions.

S3 Recovery and Handling Make Distinct Contributions

Refuse-All is a separate calibration. Write $\mathcal{H}(x, y) = 2xy/(x + y)$ for positive $x + y$, and $\mathcal{H}(0, 0) = 0$. Let x denote SafeEX and y denote SHR; subscripts 0 and 1 identify a comparator and ACCESS, respectively. We use confirmed

lower SafeEX endpoints and the resolved SHR values. The two-component attribution averages the orders in which recovery and handling are changed:

$$C_{\text{rec}} = \tfrac{1}{2}[\mathcal{H}(x_1, y_0) - \mathcal{H}(x_0, y_0) + \mathcal{H}(x_1, y_1) - \mathcal{H}(x_0, y_1)],$$
$$C_{\text{hand}} = \tfrac{1}{2}[\mathcal{H}(x_0, y_1) - \mathcal{H}(x_0, y_0) + \mathcal{H}(x_1, y_1) - \mathcal{H}(x_1, y_0)].$$

The intermediate terms cancel, giving $C_{\text{rec}} + C_{\text{hand}} = \mathcal{H}(x_1, y_1) - \mathcal{H}(x_0, y_0)$. This finite-change accounting treats the two update orders symmetrically. The table describes the observed score difference, with each row summing to its total; it is not a component-removal experiment or a confidence interval.

Table S2: BSS gain decomposition

Model	Versus	Recovery	Handling	Total
DeepSeek Flash	Direct	+17.35	+6.33	+23.68
DeepSeek Flash	Serial	-1.62	+10.65	+9.04
DeepSeek Flash	Policy	-1.92	+29.59	+27.67
Qwen2.5-14B	Direct	+12.85	+19.70	+32.55
Qwen2.5-14B	Serial	-2.38	+24.88	+22.50
Qwen2.5-14B	Policy	-2.14	+24.54	+22.40
Arctic R1-7B	Direct	+8.86	+47.68	+56.53
Arctic R1-7B	Serial	-2.31	+59.33	+57.01
Arctic R1-7B	Policy	-2.44	+25.38	+22.94
Qwen2.5-7B	Direct	+22.70	+12.59	+35.29
Qwen2.5-7B	Serial	+11.27	+10.09	+21.37
Qwen2.5-7B	Policy	+2.37	+25.07	+27.44
Qwen3.8-27B	Direct	+23.58	+2.19	+25.77
Qwen3.8-27B	Serial	-1.97	+8.39	+6.41
Qwen3.8-27B	Policy	-1.06	+28.51	+27.44

Note. **Safe handling drives most gains over interactive baselines.** BSS attribution at confirmed endpoints, ACCESS minus comparator (pp).

Population contributions remain additive within each component, where Δ means ACCESS minus the comparator:

$$\Delta\,\text{SafeEX} = \frac{|A|}{|E|}\Delta\,\text{SafeEX}_A + \frac{|P|}{|E|}\Delta\,\text{SafeEX}_P, \qquad \Delta\,\text{SHR} = \frac{|Z|}{|\mathcal{I}_{\text{den}}|}\Delta\,\text{CRR}_Z + \frac{|P|}{|\mathcal{I}_{\text{den}}|}\Delta\,\text{SHR}_P\,.$$

For DeepSeek versus Serial, the two recovery contributions are -4.80 and $+2.30$ points; the two handling contributions are $+27.97$ and $+0.64$. The harmonic score transforms these component changes into -1.62 recovery and $+10.65$ handling points. Consequently, the population proportions weight SafeEX and SHR internally, rather than serving as fixed BSS weights. Table S2 therefore attributes the DeepSeek advantage over Serial to safer handling, chiefly correct refusal on Z; the separate recovery comparisons establish the partial-service benefit.

For the disjoint-population check, use $B_{AZ} = \mathcal{H}(\text{SafeEX}_A, \text{CRR}_Z)$, excluding P from both components. ACCESS's lower endpoint exceeds each comparator's upper endpoint for every model. Section S15 reports these margins and varies the relative emphasis using weighted harmonic aggregation.

S4 Partial-Recovery Gains Survive Paired Testing

We resample the 28 source-question clusters with replacement within each data source, keeping each stratum's cluster count fixed. All associated role/specificity variants and pipeline pairs remain together. Each of 10,000 replicates (seed 20261106) recomputes the denominator after cluster resampling. The intervals below concern SafeEX_P differences. All scoring evidence on recoverable denied requests (P) is resolved; interval width here represents cluster resampling rather than unresolved permissions.

Table S3 gives positive adjusted lower endpoints in all 15 comparisons on the fixed 28-cluster population. The higher recovery on recoverable denied requests (P) therefore persists after accounting for paired source-question sampling and multiple comparisons.

Table S3: Recovery on recoverable denied requests

Model	Versus	Difference	95% interval	Adjusted interval
DeepSeek Flash	Direct	40.94	[26.20, 54.69]	[19.34, 61.77]
DeepSeek Flash	Serial	42.60	[28.95, 55.21]	[22.31, 61.89]
DeepSeek Flash	Policy	32.19	[21.13, 42.95]	[16.67, 47.62]
Qwen2.5-14B	Direct	22.19	[9.45, 36.36]	[3.63, 43.23]
Qwen2.5-14B	Serial	38.12	[24.70, 51.88]	[19.39, 59.30]
Qwen2.5-14B	Policy	31.25	[20.00, 42.11]	[14.63, 46.84]
Arctic R1-7B	Direct	29.58	[16.87, 41.88]	[10.84, 46.37]
Arctic R1-7B	Serial	33.33	[22.08, 43.90]	[17.08, 48.32]
Arctic R1-7B	Policy	30.00	[19.48, 40.51]	[14.75, 44.87]
Qwen2.5-7B	Direct	38.33	[22.92, 53.16]	[15.83, 60.78]
Qwen2.5-7B	Serial	45.00	[29.76, 59.76]	[22.78, 67.07]
Qwen2.5-7B	Policy	42.08	[26.92, 56.79]	[19.92, 63.89]
Qwen3.8-27B	Direct	43.85	[29.49, 57.38]	[22.37, 64.22]
Qwen3.8-27B	Serial	42.60	[28.85, 55.41]	[21.95, 62.30]
Qwen3.8-27B	Policy	36.25	[22.76, 49.38]	[17.48, 56.16]

Note. **All 15 adjusted partial-recovery gains remain positive.** SafeEX_P improvement in percentage points. Adjusted intervals use percentile tails $0.05/(2 \times 15)$ for the family of 15 paired comparisons.

The overall analysis resamples 476 source-question clusters within data source and recomputes SafeEX and SHR on their eligible populations before forming BSS. For unresolved recovery evidence, lower and upper difference endpoints are resampled separately; percentile envelopes include sampling and evidence uncertainty without midpoint imputation. These intervals measure source-question uncertainty conditional on the frozen references and recorded runs.

S5 Benchmark-Allowed Outcome Assignment

Overall output fulfillment depends mainly on benchmark-allowed requests. SafeEX averages fulfillment across both benchmark-allowed requests A and reference-positive benchmark-denied requests P:

$$\text{SafeEX} = \frac{|A|}{|E|}\,\text{SafeEX}_A + \frac{|P|}{|E|}\,\text{SafeEX}_P\,.$$

Requests in A account for 94.60% of the output-evaluation cohort E, whereas P accounts for only 5.40%. For DeepSeek Flash, comparing the confirmed lower endpoints, ACCESS improves SafeEX_P over Policy by 32.19 percentage points (48.75% versus 16.56%), but its SafeEX_A is 5.94 points lower (54.60% versus 60.54%). Because SafeEX weights requests equally, the improvement on the smaller subset P does not offset the decrease on A. ACCESS therefore fulfills more authorized output on benchmark-denied requests while achieving lower overall SafeEX than Policy.

We assign each allowed record to the first applicable category in this ordered list: terminal refusal; nonzero-width evidence interval; complete recovery; partial-column recovery; absent SQL without recorded refusal; confirmed unauthorized SQL; recorded execution error; row-cardinality mismatch; ordering that invalidates a positive unordered score; other zero output agreement. This precedence yields mutually exclusive categories and preserves all 1,402 instances per arm. Parse, planning, and runtime errors are included in the recorded-execution-error category. Row-count mismatch means that the candidate returns more or fewer rows than the reference, counting duplicates; equal row counts may still fail tuple-value or ordering checks. Refusal counts all recorded terminal refusals within the fixed benchmark-allowed set. These are the categories used in Table V of the main paper; fine-grained frontend causes require their corresponding trace evidence.

Table S4 records 732 complete output targets for ACCESS, versus 811 for Serial and 824 for Policy with DeepSeek Flash. These are confirmed Full_A successes; SafeEX_A additionally credits partial columns with complete matching rows.

Table S4 shows that the DeepSeek recovery gap relative to Policy combines more ACCESS refusals and row mismatches with similar execution-error counts. Joining ACCESS's pruning records by request ID identifies 201

Table S4: Outcomes on benchmark-allowed requests

Model	Pipeline	C	P	R	N	A	E	B	O	X	U
DeepSeek Flash	Direct	450	62	0	37	56	314	302	21	146	14
DeepSeek Flash	Serial	811	49	0	26	76	86	160	10	142	42
DeepSeek Flash	Policy	824	48	29	0	99	73	153	12	116	48
DeepSeek Flash	ACCESS	732	63	138	0	58	72	188	14	123	14
Qwen2.5-14B	Direct	347	43	0	22	53	482	307	8	126	14
Qwen2.5-14B	Serial	701	23	0	27	72	287	146	2	122	22
Qwen2.5-14B	Policy	687	25	29	0	86	285	158	3	105	24
Qwen2.5-14B	ACCESS	601	37	138	8	66	251	173	6	104	18
Arctic R1-7B	Direct	382	40	0	16	113	305	337	11	180	18
Arctic R1-7B	Serial	734	25	0	17	89	255	144	4	111	23
Arctic R1-7B	Policy	716	21	29	12	85	230	167	1	119	22
Arctic R1-7B	ACCESS	626	26	138	20	40	210	196	3	124	19
Qwen2.5-7B	Direct	277	31	0	64	68	561	277	9	101	14
Qwen2.5-7B	Serial	480	22	0	341	65	300	113	2	68	11
Qwen2.5-7B	Policy	635	13	29	0	103	377	124	4	95	22
Qwen2.5-7B	ACCESS	647	39	138	4	70	71	183	5	131	114
Qwen3.8-27B	Direct	367	47	0	36	58	531	215	14	119	15
Qwen3.8-27B	Serial	788	39	0	38	74	142	151	5	140	25
Qwen3.8-27B	Policy	774	33	29	3	85	182	154	7	112	23
Qwen3.8-27B	ACCESS	702	58	138	1	45	71	190	12	126	59

Note. **Refusal and row mismatch distinguish benchmark-allowed outcomes.** Disjoint allowed outcomes. C: complete; P: partial columns; R: refusal; N: absent SQL; A: unauthorized; E: execution error; B: row-count mismatch; O: order rejection; X: other zero agreement; U: inconclusive scoring evidence. Each row sums to 1,402.

benchmark-allowed cases: 138 refusals, three complete recoveries, 13 partial-column recoveries, and 47 zero-credit deliveries. Their lower score sum is $3 + 7.8333 = 10.8333$. These associations identify retained-goal decisions for semantic inspection; they do not by themselves establish whether a specific deletion was necessary.

Pruning alone does not guarantee correct output: the frontend must retain viable goals, and the model must generate correct SQL for them.

Refusal reduces completion on benchmark-allowed requests. On the fixed 1,402 requests in A, ACCESS records 138 terminal refusals (9.84%) and Policy records 29 (2.07%); Direct and Serial record none. Counts are identical across models because each pipeline replays the same frontend outcomes. The denominator is the full benchmark-allowed set used in the main comparison. These refusals account for part of the output-fulfillment gap.

S6 Remaining Score Uncertainty Is Confined to Allowed Requests

Ten previously unresolved positive-denied scores have native authorized evidence in the handling evaluation: eight Qwen2.5-7B and two Qwen3.8-27B ACCESS candidates. Reconciliation requires equality of candidate ID, request ID, source cluster, and exact SQL SHA256. Each had a conditional full-row score of one; ordered cases additionally have an independently recorded ordered match of one. Their intervals therefore change from $[0, 1]$ to $[1, 1]$. SQL, policy, reference membership, and generated output are not modified. This changes SafeEX_P to 46.25% and 48.75%, respectively, and updates all affected overall scores, plots, and bootstrap analyses.

Among ACCESS's remaining records, permission uncertainty is concentrated in LiveSQLBench (3 on Qwen2.5-14B, 100 on Qwen2.5-7B, 45 on Qwen3.8-27B). Empty-reference uncertainty is in Spider (14, 15, 18, 14, 14 across models); Arctic has one execution-pending BIRD record. Empty outputs do not identify a unique column correspondence from values, even if both queries execute. More compute is therefore not a general solution to all three uncertainty sources. Table S5 therefore places the remaining evidence bounds in allowed-request recovery; the positive-denied comparison is fully resolved.

Table S5: Unresolved evaluation evidence

Model	Pipeline	Authorization	Empty reference	Execution pending	Total
DeepSeek Flash	Direct	2	12	0	14
DeepSeek Flash	Serial	19	19	4	42
DeepSeek Flash	Policy	27	18	3	48
DeepSeek Flash	ACCESS	0	14	0	14
Qwen2.5-14B	Direct	0	12	2	14
Qwen2.5-14B	Serial	3	18	1	22
Qwen2.5-14B	Policy	6	16	2	24
Qwen2.5-14B	ACCESS	3	15	0	18
Arctic R1-7B	Direct	1	17	0	18
Arctic R1-7B	Serial	2	20	1	23
Arctic R1-7B	Policy	0	21	1	22
Arctic R1-7B	ACCESS	0	18	1	19
Qwen2.5-7B	Direct	0	12	2	14
Qwen2.5-7B	Serial	3	8	0	11
Qwen2.5-7B	Policy	5	17	0	22
Qwen2.5-7B	ACCESS	100	14	0	114
Qwen3.8-27B	Direct	1	14	0	15
Qwen3.8-27B	Serial	8	17	0	25
Qwen3.8-27B	Policy	5	18	0	23
Qwen3.8-27B	ACCESS	45	14	0	59

Note. **All remaining score uncertainty is confined to A.** Remaining unresolved SafeEX records after reconciliation. All are benchmark-allowed; P has no unresolved scores.

S7 Evaluation Implementation and Comparison Scope

Complete tuples must preserve the required order. Candidate and Gold columns are matched injectively by their observed values. Selected columns must jointly reproduce every reference row and duplicate, without extra projected rows. Native `DENSE_RANK` over the Gold sort expressions defines equal-key tie blocks. Rows may permute within a block, not across blocks. Only the final outermost `ORDER BY` constrains output order; an inner or window order alone does not. The Gold execution fixes `LIMIT/OFFSET` membership even at a tied boundary. Numeric values are normalized to eight decimal places. Authorization covers all referenced expressions, including hidden filtering, joining, and ordering dependencies.

Exact additionally requires every candidate column to match. Full requires every target column to match, while permitting additional authorized columns. Exact in Section S13 adds the native result-width condition, so every candidate column participates in the full-reference correspondence. Its native-width metadata is separate from the scorer's list of matched column indices.

Common inputs support comparisons of policy placement. The Serial/Policy frontends use DeepSeek Flash with temperature 0, a 512-token response cap, a maximum of 30 turns and 80 calls, and the same requester-simulator reference-intent protocol. Their conversational backbone adapts BIRD-INTERACT commit `451fe2c`; the target contract is an explicit natural-language request. They receive the frozen schema without external knowledge or execution feedback. Policy adds permissions and prompt-level pruning/refusal during refinement; Serial supplies permissions only at SQL generation. Direct + RBAC supplies the initial request and permissions directly to the SQL model without dialogue. Policy and ACCESS use policy-free SQL-generation prompts after permission-aware refinement. The DeepSeek Flash SQL setting is temperature 0, a 4,096-token cap, no demonstrations, and disabled thinking. All four pipelines use the same role policy for final SQL authorization. Thus the comparison shares evaluation permissions while placing policy information at the stages defined by each pipeline.

Within each pipeline, five-model evaluation replays that pipeline's same frozen frontend outcomes and refined requests. Every main-table arm shares request IDs, Gold queries, role policies, fixed output targets, and evaluation rules. Request-level frontend counters cover all three interactive pipelines. S9 specifies their record linkage; S10 specifies variant recovery and S14 reconciles the available SQL-stage usage.

Executable checks and human review establish reference quality. The evaluation combines executable reference checks, source lineage, and independent human review of texts and references. The running example in the paper illustrates the design; quantitative comparisons use the recorded candidate and trace identities.

S8 Recovery Gains Persist across Reference Subsets

Four diagnostic subsets separate reference properties. Let G_i be all output-column positions in the full Gold query. The partial-projection subset $P_{\text{part}} = \{i \in P : L_i \subsetneq G_i\}$ contains 53 requests in 18 source clusters: each reference retains a nonempty proper subset of Gold output columns. Within P_{part}, P_G contains references beyond grouping-label-only outputs (47 instances, 16 clusters), and P_S contains structurally supported partial-output witnesses (47 instances, 16 clusters). $P_{SG} = P_S \cap P_G$ contains 41 instances in 14 clusters. Both 47-instance sets are accepted reference subsets with different operator criteria; all pipelines use identical IDs within each diagnostic. These subsets describe output-reference properties.

For each subset we retain the existing ordered, full-row column scores and perform 10,000 paired, data-source-stratified source-cluster bootstrap replicates (seed 20261106). Each replicate recomputes its request denominator. The tables report the observed score and the ordinary 95% interval for ACCESS minus each comparator; they are conditional diagnostics, not independent replications or human-annotation intervals. The accompanying analysis also records 15-comparison envelopes per subset and 60-comparison envelopes across all four new subset analyses. No unadjusted interval is labeled as family-wise significance.

Table S6: Recovery on partial-projection references

Model	D	S	Pol.	ACCESS	vs. D	vs. S	vs. Pol.
DeepSeek Flash	5.66	0.00	7.55	52.83	[27.8, 65.4]	[35.2, 69.4]	[29.6, 60.8]
Qwen2.5-14B	28.30	1.89	3.77	50.94	[7.4, 39.6]	[31.5, 65.4]	[29.6, 63.5]
Arctic R1-7B	5.66	0.00	5.66	50.94	[25.9, 64.2]	[34.0, 66.7]	[29.6, 61.1]
Qwen2.5-7B	9.43	0.00	3.77	52.83	[26.4, 59.6]	[35.2, 69.4]	[32.1, 65.4]
Qwen3.8-27B	0.00	0.00	9.43	52.83	[35.2, 69.4]	[35.2, 69.4]	[27.8, 59.3]

Note. **ACCESS leads recovery on partial-output references.** P_{part}: proper subsets of Gold output columns, $N = 53$. SafeEX scores (%) and ACCESS-minus-comparator 95% intervals (pp).

Table S7: Recovery without grouping-only outputs

Model	D	S	Pol.	ACCESS	vs. D	vs. S	vs. Pol.
DeepSeek Flash	6.38	0.00	8.51	59.57	[33.3, 70.2]	[43.5, 73.9]	[35.4, 66.0]
Qwen2.5-14B	31.91	2.13	4.26	57.45	[8.3, 43.5]	[38.3, 70.8]	[37.5, 68.1]
Arctic R1-7B	6.38	0.00	6.38	57.45	[31.2, 68.8]	[41.7, 71.7]	[35.4, 66.0]
Qwen2.5-7B	10.64	0.00	4.26	59.57	[31.9, 64.6]	[43.5, 73.9]	[39.6, 70.2]
Qwen3.8-27B	0.00	0.00	10.64	59.57	[43.5, 73.9]	[43.5, 73.9]	[33.3, 63.8]

Note. **ACCESS's recovery lead persists beyond grouping-label outputs.** P_G: grouping-label cases excluded, $N = 47$. SafeEX scores (%) and ACCESS-minus-comparator 95% intervals (pp).

Table S8: Recovery on structurally supported targets

Model	D	S	Pol.	ACCESS	vs. D	vs. S	vs. Pol.
DeepSeek Flash	6.38	0.00	8.51	55.32	[27.7, 68.1]	[37.5, 72.3]	[30.4, 63.0]
Qwen2.5-14B	29.79	2.13	4.26	51.06	[8.3, 35.6]	[31.2, 66.0]	[30.4, 63.0]
Arctic R1-7B	6.38	0.00	6.38	53.19	[25.5, 66.7]	[36.2, 69.6]	[31.1, 63.0]
Qwen2.5-7B	10.64	0.00	4.26	55.32	[27.1, 62.2]	[37.5, 72.3]	[33.3, 68.1]
Qwen3.8-27B	0.00	0.00	8.51	55.32	[37.5, 72.3]	[37.5, 72.3]	[30.4, 63.0]

Note. **ACCESS leads recovery on structurally supported witnesses.** P_S: structurally supported witnesses, $N = 47$. SafeEX scores (%) and ACCESS-minus-comparator 95% intervals (pp).

Table S9: Recovery with both reference restrictions

Model	D	S	Pol.	ACCESS	vs. D	vs. S	vs. Pol.
DeepSeek Flash	7.32	0.00	9.76	63.41	[35.0, 73.8]	[47.6, 76.9]	[38.1, 68.3]
Qwen2.5-14B	34.15	2.44	4.88	58.54	[9.5, 40.0]	[40.5, 70.7]	[39.0, 67.5]
Arctic R1-7B	7.32	0.00	7.32	60.98	[31.7, 72.5]	[45.2, 73.8]	[38.1, 68.3]
Qwen2.5-7B	12.20	0.00	4.88	63.41	[34.1, 67.5]	[47.6, 76.9]	[42.9, 73.2]
Qwen3.8-27B	0.00	0.00	9.76	63.41	[47.6, 76.9]	[47.6, 76.9]	[38.1, 68.3]

Note. **The recovery lead persists under both reference exclusions.** P_{SG}: both exclusions, $N = 41$. SafeEX scores (%) and ACCESS-minus-comparator 95% intervals (pp).

Tables S6–S9 show ACCESS with the highest recovery score in every model under each reference restriction. All scores in these subsets are resolved. On partial-projection references, ACCESS scores 50.94–52.83% across models. The 58.54–63.41% range under both structural and grouping exclusions likewise describes variation across models, not unresolved evidence. The recovery advantage therefore persists for strict partial-output targets and beyond grouping-label-only outputs. These diagnostics retain candidate SQL and vary only the reference category; Table S8's structurally supported witnesses additionally preserve duplicate behavior through the retained query structure.

Higher-replicate checks preserve the structural-subset gains. The tail-stability check cited in Section V-F of the main paper repeats the paired, data-source-stratified cluster bootstrap with 200,000 replicates for each of three seeds: 20261106, 20261107, and 20261108. It uses the same four reference subsets and the same 60-contrast family adjustment, with percentile tails $0.05/(2 \times 60)$ and $1 - 0.05/(2 \times 60)$. For all five ACCESS-minus-Policy contrasts on P_S summarized in the main paper's Table VIII, the adjusted lower endpoint remains positive in every run. These additional runs check the stability of the reported comparison directions; the main table retains the original 10,000-replicate intervals.

S9 Frontend Configuration, State Contracts, and Interaction Records

Record identities align the compared populations. All three interactive evaluation files contain the same 2,800 unique IDs. For ACCESS, every stored trace SHA256 and terminal outcome agrees with the original main-experiment pruning ledger and historical trace manifest. For Serial/Policy, request IDs, normalized trace basenames, and outcomes agree with their corresponding manifests. This reanalysis verifies exported record identity; the original historical raw traces are not re-read. ACCESS's 3,016 user interactions come from the archived 2,800-trace audit, while Serial/Policy interaction totals come from per-request fields.

Table S10: Recorded interaction cost

Counter	Serial	Policy	ACCESS
User interactions, total	11,810	15,014	3,016
Logical calls, total	26,466	32,887	19,218
Logical calls, mean	9.45	11.75	6.86
Logical calls, median	7	7	5
Input tokens, total	91,075,986	135,317,418	72,198,669
Output tokens, total	1,505,248	2,017,583	3,559,259
Total tokens, mean	33,064.73	49,048.21	27,056.40
Total tokens, median	12,641.50	14,718.50	19,408.00

Note. **ACCESS uses fewer interactions and lower mean frontend tokens.** Recorded frontend counters for the shared population. Tokens exclude downstream SQL generation; values are not monetary cost or complete retry-inclusive billing.

ACCESS has lower mean total tokens but higher median total tokens and mean output tokens than either interactive comparator. These are unconditional averages including refusals; early termination and different delivered-service rates affect them. The field `logical_calls` is reported as recorded, without reinterpreting it as the number of paid API attempts. The main-run counters do not establish complete transport-retry billing. For the separate ablation traces, Section S10 reconciles logical reuse with actual recorded usage. Direct has no clarification frontend; its SQL cost is outside this table, as are all three interactive pipelines' downstream SQL costs. Table S10 establishes lower

average frontend effort; Tables S17 and S18 test whether that saving remains after conditioning on outcomes and successful service.

Generation snapshots specify the workflow configuration. DeepSeek Flash is the display name used in the main paper; the recorded API model identifier in the inspected manifests is `deepseek-flash`. Main ACCESS outcomes are identified by the `v10_allcase_shard` trace manifest. The `v10_full` executable snapshot records the `deepseek-flash` alias, temperature zero, a common recorder for decomposition/refinement/review and requester calls, 30 turns, an 80-attempt shared budget, and `requester_protocol_v6_one_stage_gold`. The main trace manifest and executable snapshot have separate identities. Module variants use the frozen workspace and serial state-repair snapshot described in S10.

Serial/Policy configuration manifests record `deepseek-flash`, temperature zero, a 512-token response cap, 30 turns, 80 calls, shared runtime hashes, and the same named requester protocol. Upstream and adapted prompt hashes accompany these manifests. The artifact identifies a provider alias, a source snapshot, and a selected-trace manifest separately so that scoring and generation records can be reconstructed at their recorded granularity.

Identity checks link candidates to common scoring inputs. Request IDs and SQL hashes join selected candidates to effective policies and reference columns. The evaluator reconstructs authorization, tuple-wise execution agreement, order-aware SafeEX, SHR, BSS, and paired statistics. Frozen frontend sources provide the operation contracts and prompts; experiment manifests associate selected outcomes with their generation and recovery stages.

Persistent state connects clarification and deferred pruning. Following Section III and Fig. 2 of the main paper, the interaction state $X_t = (G_t, Q_t, O_t, B_t)$ maintains the requirement graph, question ledger, obligation pool, and deferred-blocking region. The graph records goals and necessary conditions; the ledger links questions and answers to their requirements; the obligation pool tracks contributions still needing resolution; and the blocking region stores proposed removals with their causes. These records let new answers update requirements and let access conflicts redirect later clarification. Dependency-aware review then checks how a removal affects surviving goals.

In Fig. 2 of the main paper, the label S denotes the schema; D, Π, and r denote the database, policy, and role. The diagram writes the state as $X = (G, Q, O, B)$ and omits the refinement-step subscripts.

State contracts separate semantic proposals from structural checks. The frozen implementation distinguishes output and modifier nodes, each with an identifier, parent lineage, business text, column bindings, and pending questions. Necessary dependencies use consumer-to-provider `requires` edges; annotation attachments use `applies_to`. A replacement maps an edited parent to one or more children. Existing attachments and contribution identities are transferred by code; new edges must involve a local replacement. Shared conditions are preserved by their identities and consumers, rather than by matching column names. The validator requires pending decisions to be routed, prior dependencies to be preserved or explicitly accounted for, and modifiers to retain an output consumer. These contracts check the submitted state; semantic proposals supply the requirement meanings and dependency classification.

Pruning review compares the residual with intent evidence. The review compares requirements before and after pruning using their nodes and edges, computed permission facts, proposed deletions, the original request, actual answers, evidence quotations, and candidate handoff text. It accepts or reports unsupported dependencies, unrepresented intent, answer conflicts, or residual scope changes. Validated issues name affected nodes and cite supplied evidence. A rejected review rolls back the deletion and returns unresolved. Ordinary unpruned handoffs do not acquire an additional global model audit from this pruning-specific check. Native SQL authorization separately tests actual query accesses; target matching supplies the execution-based recovery evidence.

A rejected clarification menu is answer evidence, rather than an SQL authorization verdict. The answer, the proposed removal, and the final recorded outcome are separate stages; the evaluation uses the recorded terminal decision and independently checks any generated SQL.

Failed local checks prevent an unvalidated handoff. In the frozen module workspace, contract validation permits up to two calls for shape/state repair. Transport exceptions are separate execution failures. For free-text answers, `answer_contribution_review` checks whether substantive answer content and necessary operands appear in the proposed binding claims. A negative coverage review rolls back the proposal and marks the attempt unresolved; it does not silently ask the user again. A final transition-audit failure also marks the attempt unresolved. Whole-attempt retries reported in Section S10 belong to the experimental runner, not to this internal scheduling step.

Reference-assisted answers supply a common latent intent. The requester receives the reference question, intent fragments, Gold SQL, the current question, and dialogue history as private inputs. Its answer is passed to the refinement agent. Construction and requester both use fragment-based latent intent. ACCESS reconstructs its own state from the public request and received answers; it is not initialized with the construction fragments. Thus direct access to private references is separated from the shared representational assumptions. The controlled comparison holds the requester protocol fixed; model replay varies realization after the frontend decisions.

Human review accepts 117 of 120 sampled texts. Two human reviewers independently assessed all 120 sampled L0/L1 texts from 60 source questions, with 20 source questions per data source and one L0/L1 pair per source question. The final human assessment accepted 117/120 texts (97.5%). The denominator includes every reviewed text, including the three not accepted. These three texts were retained in the original experiments and remain included in the reported results. Keeping them preserves the evaluated sample rather than selectively excluding reviewed failures. This is the final acceptance rate for the sampled rewrites, not inter-rater agreement, model execution accuracy, or an estimate of the quality of all legal-output references. The sample was selected after score analysis; paired rewrites from one source question are not independent source-question observations.

Human review confirms all 80 references and the negative cohort. Two human reviewers independently assessed all 80 references under the evaluation contract: authorization, preservation of required result tuples and duplicate multiplicity, and final ordering on the benchmark database. A final consolidated assessment accepted all 80 references (100%). Two human reviewers also independently examined every one of the 1,318 requests in Z; both found no additional legal service satisfying the original requirements.

S10 Ablation Protocol and Recovery

The three completed variants intervene on the ACCESS frontend: w/o clarification suppresses user/simulator interaction but retains internal analysis and permission-based pruning; w/o selective pruning retains clarification and actual permissions, but replaces selective removal with whole-request refusal; w/o coordination completes clarification first and prohibits further user questions during the pruning phase. The policy remains available in all three variants. As in Section V-B of the main paper, the coordinated and sequential workflows both retain persistent state; w/o coordination changes when clarification can occur, completing it before the pruning phase. The comparison examines how clarification and selective pruning use the shared state together. This sequential variant differs from the main Serial baseline, which introduces policy only at SQL generation.

All variants retain the same request population. Each variant contains the identical 2,800 request IDs, including all refusals. SQL is generated only for terminal deliveries. Every selected trace hash is verified against its selection manifest and retry audit, including identity, terminal outcome, retained policy, and module flags. Full ACCESS reuses the unchanged historical DeepSeek Flash main run, keeping its candidates and outcomes identical to the main comparison. Each variant supplies its recorded module condition under the recovery protocol below. The variants' SQL configuration uses the recorded `deepseek-flash` alias, temperature zero, a 4,096-token cap, thinking disabled, no demonstrations, and no policy in the SQL prompt. A service alias does not establish an identical immutable provider revision across dates.

Terminal selection is independent of SQL correctness. Selection takes each request's first genuine `deliver`/`refuse` in attempt order, independently of SQL correctness or Gold scores. Failed attempts remain in the effort audit. Recovery includes repeated inputs and validator-error feedback, which can change initial inputs or contract-repair prompts. Feedback uses the public request, prior response, validator diagnostics, and confirmed user choices; it excludes Gold and SQL-evaluation feedback. Late sequential recovery extends contract-repair attempts under unchanged validators. The comparison evaluates completed workflows with this recorded recovery effort.

The sequential variant uses a corrected state-handoff implementation throughout all 2,800 requests; successes from the superseded implementation are not mixed in. Twenty later terminal reruns caused by interrupted scheduling cannot replace earlier selected outcomes, but remain in recorded effort. Six interrupted calls have no usage record; their missing cost is unknown, not zero. Sequential effort below excludes the separately archived superseded implementation.

Table S11 shows terminal outcomes for all 2,800 requests in every variant, despite initial completion of 90.54–94.61%. These final scores therefore represent completed workflows with recovery, and the sequential variant incurs the most recorded effort. Recorded tokens include failed attempts and available interrupted-call usage, but exclude

Table S11: Ablation completion and cost

Quantity	w/o clarification	w/o selective pruning	w/o coordination
Final deliveries / refusals	1,806 / 994	1,294 / 1,506	1,478 / 1,322
Initial terminal outcomes	2,631	2,649	2,535
Initial completion (%)	93.96	94.61	90.54
Recorded attempts	3,194	3,232	3,549
Selected traces using feedback	2	5	2,440
All-attempt logical calls	10,900	16,069	30,783
All-attempt tokens	47,481,243	61,366,269	137,127,210
Tokens per fixed request	16,957.59	21,916.52	48,974.00
Interrupted calls, usage unknown	0	0	6

Note. **All variants reach terminal outcomes, with different recovery effort.** Completion and recorded frontend effort. Initial completion is the number of requests reaching a genuine terminal outcome on their earliest recorded attempt, divided by 2,800. Feedback counts selected traces that use validator-error diagnostics for contract repair, including traces first attempted after feedback tooling was enabled. These counts are neither user-interaction totals nor counts of requests requiring post-failure retries.

SQL generation and missing usage. Historical full-arm counters in Section S9 have a different retry-coverage boundary and cannot establish a matched-cost advantage over these totals.

Actual usage excludes reused responses. A trace's `cost` includes logical reuse, whereas `actual_cost` excludes responses reused without another model call. Rechecking 149 selected traces (two w/o selective pruning and 147 sequential) matches their frozen hashes and accounts for every difference using `reused_from` and recorded usage. The following table uses actual usage for selected attempts and the same accounting for all attempts. The additional percentage is (all-attempt tokens/selected-attempt tokens $-1) \times 100$; it includes failed attempts and recorded post-selection reruns, not solely retry overhead.

Table S12: Ablation token usage

	w/o clarification	w/o selective pruning	w/o coordination
Selected actual tokens (M)	40.078	51.784	91.495
All-attempt tokens (M)	47.481	61.366	137.127
Additional effort (%)	18.47	18.50	49.87

Note. **Counting all attempts reveals additional recorded effort.** Selected and all-attempt frontend tokens. M denotes millions. All-attempt totals exclude missing usage and SQL generation.

Table S12 shows 18.47%, 18.50%, and 49.87% more recorded tokens when all attempts are counted for w/o clarification, w/o selective pruning, and w/o coordination. Successful-attempt totals therefore understate the effort spent reaching these terminal results, especially for the sequential workflow. All totals are recomputed from the 2,800 request-level attempt audits for each variant. Selection matches the first genuine terminal trace by hash, rather than the most accurate SQL. The sequential total includes 20 later terminal reruns and excludes the superseded implementation; six interrupted calls lack usage. Logical model-use counters, actual recorded usage, and a complete provider bill are therefore distinct. No dollar cost or end-to-end latency is inferred. Original SQL, manifests, attempt audits, and source hashes remain in the evidence archive.

S11 Full ACCESS Recovers More Restricted Legal Output

All ablation results in this supplement use the paper's order-aware SafeEX and harmonic BSS. Existing candidates are rescored on the same fixed $|E| = 1,482$ and $|P| = 80$ as the main table, with request-macro column fractions, complete result tuples and duplicate multiplicity, native whole-SQL authorization, and final Gold ordering. Refusals retain their denominators. Extra authorized output columns follow the same recall-like treatment in every arm.

Reference IDs, policies, target columns, and selected Gold alternatives match the main reference. Native checks reconcile unresolved candidate-access records and all 240 P instances across the three variants. The same order-aware evaluator performs 320 nontrivial candidate checks across 43 source clusters. The resulting 4,446 recovery records and 4,194 handling records are joined to unchanged historical full-arm records. Original source hashes are unchanged after rescoring. SQL candidates and output targets remain fixed.

Table S13: Ablation results

Variant	BSS	SafeEX	SHR	SafeEX$_A$	SafeEX$_P$	SafeEX$_{P_S}$
ACCESS (full)	67.23–67.95	54.28–55.23	88.27	54.60–55.60	48.75	55.32
w/o clarification	44.92–45.65	33.28–34.09	69.10	33.46–34.32	30.00	27.66
w/o selective pruning	66.87–67.88	51.74–52.96	94.49	53.77–55.05	16.25	6.38
w/o coordination	66.73–67.51	53.54–54.55	88.56	54.60–55.67	35.00	46.81

Note. **Full ACCESS recovers more denied-request output than each ablation.** Module-variant scores (%). P contains 80 recoverable denied requests; P_S contains 47 structurally supported partial references. All P and P_S scores are resolved. w/o coordination retains clarification and pruning in a fixed sequential schedule.

Table S14: Ablation handling and recovery

Variant	Z refuse	P refuse	P safe SQL	P other	Full$_A$	Full$_P$	Full$_{P_S}$
ACCESS (full)	1,160	15	59	6	732	39	26
w/o clarification	912	8	46	26	434	24	13
w/o selective pruning	1,241	58	22	0	731	13	3
w/o coordination	1,166	18	54	8	735	28	22

Note. **Selective pruning preserves service that whole-request refusal discards.** Handling and complete recovery. $Z/P/P_S$ contain 1,318/80/47 requests. Z non-refusals are implicit; the three P handling columns partition all 80 requests. Full counts are confirmed lower endpoints.

In these four DeepSeek arms, the P scores happen to be binary: positive credits recover every legal column. The metric nevertheless remains fractional-column SafeEX. Only A retains evidence uncertainty: full ACCESS has 14 empty-reference cases; w/o clarification has 12; w/o selective pruning has 17 empty-reference cases and one execution-pending case; w/o coordination has 15 empty-reference cases. These produce evidence bounds, not sampling intervals.

Tables S13 and S14 show 58 refusals and 22 safe deliveries on recoverable denied requests (P) without pruning, versus 15 and 59 for full ACCESS; complete recoveries rise from 13 to 39 with the full workflow. Selective pruning therefore preserves service that whole-request refusal often discards. Authorized partial SQL can earn credit in every arm, even without an internal pruning event; the higher score reflects recovered output rather than declarations of pruning.

Table S13 gives ACCESS the highest confirmed BSS endpoint, 67.23%, versus 66.87% for w/o selective pruning and 66.73% for w/o coordination, with overlapping evidence ranges. Its SafeEX$_P$ is 48.75%, versus 30.00%, 16.25%, and 35.00% for the three variants. These component scores locate the full workflow's clearest advantage in recovering restricted legal output; Section S12 evaluates the corresponding paired differences. In particular, the coordinated workflow recovers 39 complete references on P, compared with 28 for the sequential workflow, giving the 13.75-percentage-point gain reported in Section V-B of the main paper. The shared state carries answers, dependencies, and removal evidence through both workflows; coordination permits these updates to affect subsequent questions and pruning decisions.

S12 Adjusted Partial-Recovery Gains Support the Full Workflow

Contrasts below are full ACCESS minus each variant in percentage points. We resample source-question clusters within datasets, paired across arms, using 10,000 replicates and seed 20261106. Overall and A analyses use 476 clusters; P uses 28 and P_S uses 16. Each replicate recomputes eligible denominators; BSS combines recovery and handling from that same replicate. Unresolved evidence widens differences by subtracting opposite endpoints. Intervals condition on selected completed workflows and fixed references, not run, recovery-protocol, or annotation uncertainty.

Ordinary intervals use 2.5% and 97.5% percentile endpoints. Family-adjusted intervals use $0.05/(2 \times 3)$ and $1 - 0.05/(2 \times 3)$ tails for three comparisons within each metric/population. These Bonferroni percentile intervals are recomputed for the paper's metric, not the legacy report's Holm tests. The adjustment does not treat all exploratory populations and metrics as one confirmatory family.

Table S15: Paired ablation contrasts

Population	Metric	Variant	Ordinary 95%	Family-adjusted
E	SafeEX	w/o clarification	[17.56, 24.50]	[16.92, 25.10]
E	SafeEX	w/o selective pruning	[-0.51, 5.24]	[-0.96, 5.61]
E	SafeEX	w/o coordination	[-1.96, 3.35]	[-2.30, 3.74]
$\mathcal{I}_{\text{den}}$	SHR	w/o clarification	[16.76, 21.61]	[16.21, 22.21]
$\mathcal{I}_{\text{den}}$	SHR	w/o selective pruning	[-8.07, -4.46]	[-8.51, -4.08]
$\mathcal{I}_{\text{den}}$	SHR	w/o coordination	[-1.57, 1.00]	[-1.86, 1.36]
$E, \mathcal{I}_{\text{den}}$	BSS	w/o clarification	[19.35, 25.17]	[18.86, 25.71]
$E, \mathcal{I}_{\text{den}}$	BSS	w/o selective pruning	[-2.18, 2.61]	[-2.56, 2.96]
$E, \mathcal{I}_{\text{den}}$	BSS	w/o coordination	[-1.65, 2.65]	[-1.92, 2.97]
A	SafeEX	w/o clarification	[17.51, 24.75]	[16.94, 25.36]
A	SafeEX	w/o selective pruning	[-2.07, 3.40]	[-2.47, 3.73]
A	SafeEX	w/o coordination	[-2.63, 2.52]	[-2.99, 2.88]
P	SafeEX	w/o clarification	[10.84, 27.16]	[8.86, 28.75]
P	SafeEX	w/o selective pruning	[20.00, 44.74]	[17.28, 47.50]
P	SafeEX	w/o coordination	[3.90, 24.10]	[2.47, 26.32]
P_S	SafeEX	w/o clarification	[16.67, 38.64]	[14.58, 41.30]
P_S	SafeEX	w/o selective pruning	[30.43, 66.67]	[26.09, 70.21]
P_S	SafeEX	w/o coordination	[-2.13, 21.74]	[-4.26, 25.00]

Note. **All three adjusted gains on recoverable denied requests are positive.** Paired cluster intervals for full ACCESS minus each variant (pp); P denotes the 80 recoverable denied requests. Sampling intervals are widened for unresolved evidence.

Table S15 shows that full ACCESS improves overall SafeEX, SHR, and BSS over w/o clarification under both interval conventions, and improves P recovery over all three variants after adjustment. On P_S, its adjusted recovery gains are positive against w/o clarification and w/o selective pruning. The adjusted BSS intervals against w/o selective pruning and w/o coordination are $[-2.56, 2.96]$ and $[-1.92, 2.97]$, respectively. The partial-recovery comparisons therefore identify the benefit more precisely than these aggregate contrasts. As reported in Section V-B, the adjusted coordination gain on P is $[2.47, 26.32]$ percentage points; on the structural subset P_S, the corresponding interval is $[-4.26, 25.00]$. The subset-specific intervals describe their respective populations.

S13 Exact Output Matching Preserves the Partial-Service Gain

For each request, let m_i be the candidate output-column count and L_i the fixed target-column set. Exact adds $m_i = |L_i|$ to the full-reference condition $c_i = 1$. All candidate columns must therefore participate in the injective reference match, retaining complete result tuples, duplicate multiplicity, and Gold final ordering. The same whole-SQL authorization gate applies.

Native result widths are matched by the exact dataset/database/SQL key. All 2,879 confirmed full recoveries across the four DeepSeek arms have resolved native width. The diagnostic reuses the latest ordered scores and changes no generated SQL.

Table S16 shows that ACCESS retains 38 of 39 complete positive-denied recoveries under exact columns; the corresponding counts for Direct, Serial, and Policy are 2/6, 4/4, and 10/13. Across E, additional-column cases account for 25/771 ACCESS full recoveries and 43/837 Policy full recoveries. The partial-service ranking persists under the exact-output criterion.

Table S16: Exact output recovery

Pipeline	Cohort	N	Full count	Exact count	Extra
Direct	E	1482	456–470	384–396	72
Direct	A	1402	450–464	382–394	68
Direct	P	80	6	2	4
Serial	E	1482	815–852	768–801	47
Serial	A	1402	811–848	764–797	47
Serial	P	80	4	4	0
Policy	E	1482	837–880	794–835	43
Policy	A	1402	824–867	784–825	40
Policy	P	80	13	10	3
ACCESS	E	1482	771–785	746–760	25
ACCESS	A	1402	732–746	708–722	24
ACCESS	P	80	39	38	1

Note. **The partial-service gain survives exact output matching.** DeepSeek exact outputs. Counts use fixed cohorts; ranges reflect unresolved score evidence. Extra counts confirmed full recoveries with additional columns; unresolved records are excluded from that count.

S14 Frontend Savings Persist per Delivered and Successful Request

Frontend usage includes refinement and requester calls. The following means condition on the recorded frontend terminal state; every workflow contains 2,800 requests. Handoffs can subsequently produce SQL, SQL-stage refusal, or generation failure. A dash denotes an empty frontend outcome group.

Table S17: Cost by terminal outcome

Pipeline	Outcome	Requests	Mean tokens	Median tokens
ACCESS	deliver	1487	29.62	19.27
ACCESS	refuse	1313	24.15	19.46
ACCESS	all	2800	27.06	19.41
Serial	deliver	2800	33.06	12.64
Serial	refuse	0	–	–
Serial	all	2800	33.06	12.64
Policy	deliver	2381	48.89	14.13
Policy	refuse	419	49.94	30.07
Policy	all	2800	49.05	14.72

Note. **Lower frontend effort also holds among handoffs.** Frontend effort by terminal outcome. Tokens are in thousands.

Table S17 shows ACCESS using 29.62k frontend tokens per handoff, compared with 33.06k for Serial and 48.89k for Policy. Lower effort therefore persists among requests actually handed to SQL generation, rather than arising only from a higher refusal share.

Successful-service cost includes failed and refused requests. For complete recovery, divide all 2,800 frontend token totals by the number of confirmed Full_E requests. For positive partial service, divide the frontend tokens for all 80 P requests by the number with $c_i > 0$. Failed and refused requests remain in the relevant effort pool. These ratios measure the effort spent per successful result rather than the mean cost among successful sessions.

Table S18: Cost per successful service

Pipeline	Full_E count	Tokens/Full_E	P positive	P tokens/positive
ACCESS	771	98.26	39	87.27
Serial	815	113.60	6	676.52
Policy	837	164.08	14	421.06

Note. **ACCESS spends fewer frontend tokens per successful service.** Effort per confirmed success. Tokens are in thousands.

Table S18 gives ACCESS the lowest frontend cost per complete recovery (98.26k tokens) and per positive recovery on P (87.27k), charging recorded frontend tokens for failed and refused requests to the same cost.

Stage coverage determines which token totals can be combined. Archived provider responses record 9.647M SQL-generation tokens for Serial (2,801 responses for 2,800 final requests, including one additional response) and 7.026M for Policy (2,381 responses). Together with selected frontend counters, the observed stage totals are 102.228M and 144.361M tokens. These sums identify the recorded stages; historical frontend transport retries and end-to-end latency are outside their telemetry.

The sequential module variant records 4.682M SQL tokens for all 1,478 deliveries. Added to 137.127M retry-inclusive frontend tokens, this gives 141.809M recorded tokens across both stages, or 50.646k per fixed request. Six interrupted frontend calls have no usage. Main ACCESS retains its SQL outputs without the corresponding provider usage log; w/o clarification and w/o selective pruning archives retain 78 and 84 newly generated responses alongside reused candidates. Their SQL-stage coverage is therefore reported separately from the complete frontend accounting in S10. The separate Direct run covers a larger generation population and is not pooled into this 2,800-request cost table.

Table VI of the main paper uses the common selected-frontend accounting for all three interactive pipelines. Reuse-aware all-attempt costs and SQL-response totals retain their distinct coverage, and recorded SQL execution seconds remain separate from inference latency.

S15 BSS Gains Survive Paired Testing and Weight Checks

BSS is the harmonic mean throughout the main paper and supplement. Each of 10,000 paired, data-source-stratified cluster replicates first computes SafeEX and SHR on their eligible request populations and then evaluates $\mathcal{H}(\text{SafeEX}, \text{SHR})$. Bounds propagate monotonically through $\mathcal{H}$. The table reports the lower-tail percentile of the lower difference and the upper-tail percentile of the upper difference. Ordinary tails are 0.025 and 0.975; family-adjusted tails are $0.05/(2 \times 15)$ and $1 - 0.05/(2 \times 15)$.

Table S19: Paired BSS gains

Model	Versus	Ordinary 95%	Family-adjusted
DeepSeek Flash	Direct	[19.92, 27.22]	[18.47, 28.48]
DeepSeek Flash	Serial	[4.81, 12.50]	[3.32, 13.76]
DeepSeek Flash	Policy	[23.39, 32.03]	[21.33, 33.73]
Qwen2.5-14B	Direct	[28.84, 36.77]	[27.27, 38.33]
Qwen2.5-14B	Serial	[18.61, 27.01]	[16.94, 28.64]
Qwen2.5-14B	Policy	[18.58, 26.84]	[16.93, 28.39]
Arctic R1-7B	Direct	[53.13, 60.85]	[51.38, 62.30]
Arctic R1-7B	Serial	[53.59, 61.33]	[51.73, 62.75]
Arctic R1-7B	Policy	[19.01, 27.59]	[17.44, 29.31]
Qwen2.5-7B	Direct	[31.60, 43.38]	[30.21, 44.88]
Qwen2.5-7B	Serial	[17.88, 29.63]	[16.46, 31.00]
Qwen2.5-7B	Policy	[23.62, 36.14]	[21.89, 37.85]
Qwen3.8-27B	Direct	[21.59, 31.50]	[19.87, 33.09]
Qwen3.8-27B	Serial	[2.78, 11.88]	[1.37, 13.09]
Qwen3.8-27B	Policy	[23.55, 33.82]	[21.83, 35.46]

Note. **All 15 adjusted BSS gains remain positive.** Paired BSS improvements, ACCESS minus each main comparator (pp).

Table S19 shows positive family-adjusted lower endpoints for all 15 BSS gains. ACCESS's combined-service advantage therefore survives paired sampling uncertainty and multiple-comparison correction across all five models. The computation preserves the same source clusters, sampled multiplicities, candidates, and component scores as the accompanying SafeEX and SHR analyses.

Let $x = \text{SafeEX}$ and $y = \text{SHR}$. For recovery weight $\alpha \in [0, 1]$ and $x + y > 0$, the weighted harmonic family is

$$\mathcal{H}_\alpha(x, y) = \frac{xy}{\alpha y + (1 - \alpha)x}, \quad 0 < \alpha < 1; \qquad \mathcal{H}_0(x, y) = y, \quad \mathcal{H}_1(x, y) = x.$$

For $0 < \alpha < 1$, the score is zero when either component is zero, including $x = y = 0$; BSS is $\mathcal{H}_{1/2}(x, y)$. Larger α places more emphasis on recovery. For each grid value, the table gives the minimum margin over all 15 comparisons

after subtracting each comparator's upper evidence endpoint from ACCESS's lower endpoint. These are descriptive evidence margins, not bootstrap intervals.

Table S20: BSS sensitivity to weighting

α	Minimum margin
0.0	+10.52
0.1	+16.68
0.2	+15.21
0.3	+11.52
0.4	+8.30
0.5	+5.46
0.6	+2.97
0.7	+0.75
0.8	-1.22
0.9	-3.09
1.0	-7.09

Model	Disjoint B_{AZ} margin
DeepSeek Flash	+7.00
Qwen2.5-14B	+22.20
Arctic R1-7B	+22.90
Qwen2.5-7B	+19.51
Qwen3.8-27B	+4.38

Note. **The BSS lead persists through $\alpha = 0.7$ and without cohort overlap.** Conservative BSS margins under weighted emphasis and disjoint populations (pp).

Table S20 shows that the ranking holds across all five models through $\alpha = 0.7$ on the evaluated grid and also when P is removed from both components. At higher recovery emphasis, some comparator evidence ranges reach or exceed ACCESS. The main BSS uses equal harmonic emphasis, with both component scores reported alongside it.

S16 Planned-Population Completion

From 591 reviewed source questions, 476 have both a benchmark-allowed and a benchmark-denied role. Two roles and three specificity levels produce the 2,856 planned instances. The main paper reports the shared 2,800-request terminal cohort; the construction counts, nonterminal outcomes, and all-planned bounds are documented here.

The shared service cohort consists of terminal ACCESS outcomes. The original completion ledger covers all 2,856 planned requests: 1,487 handoffs, 1,313 refusals, 55 unresolved attempts, and one system error. The main comparison uses the same 2,800 terminal IDs for every pipeline. Refusals are included; the 56 nonterminal attempts are excluded from every pipeline's paired service score. This gives matched IDs conditional on ACCESS completion. The planned-population completion rate is reported separately and does not establish random missingness.

Table S21: Planned frontend completion

Source	Planned	Handoff	Refusal	Unresolved	System error
BIRD	1,026	515	501	9	1
Spider	1,128	605	506	17	0
LiveSQLBench	702	367	306	29	0
Total	2,856	1,487	1,313	55	1

Note. **Nonterminal attempts concentrate more in LiveSQLBench.** Terminal outcomes are handoff or refusal; unresolved and system-error attempts are not refusals. The table uses recorded frontend outcomes before SQL generation.

Table S21 gives nonterminal rates of 0.97%, 1.51%, and 4.13% for BIRD, Spider, and LiveSQLBench, respectively. By specificity, L0/L1/L2 have 18/21/17 nonterminal instances out of 952 each. By benchmark permission, the counts are 26 allowed and 30 denied out of 1,428 each. The aggregate ledger establishes these outcome categories; it does not separate dependency, contract, and budget causes within the 55 unresolved records. Query-difficulty effects require that finer trace evidence.

Worst-case completion credit preserves the BSS ordering. We bound the possible effect of the 56 excluded requests without selecting targets from candidate outputs. Let $m_A = 26$ and $m_D = 30$ be their benchmark-allowed and benchmark-denied counts, and let $k \in \{0, \ldots, m_D\}$ denote the number of recoverable denied requests among them. The all-planned denominators would be $|E| + m_A + k$ for SafeEX and $|\mathcal{I}_{\text{den}}| + m_D$ for SHR. Existing targets and per-request scores remain fixed. All excluded ACCESS attempts are unresolved or system errors, so they

receive zero recovery and handling credit. For each comparator, we allow score one on every added recovery target and handling credit one on every added denied request.

Let U^{-}_{ACCESS} be ACCESS's existing recovery-score sum at lower endpoints and V_{ACCESS} its resolved handling-credit sum. For comparator j, let U^{+}_{j} and V_j be the corresponding upper recovery and resolved handling sums. Using the harmonic mean $\mathcal{H}$ from the main paper, let $\Delta_j(k)$ be ACCESS minus comparator j BSS on this expanded cohort. It obeys

$$\Delta_j(k) \;\geq \mathcal{H}\left(\frac{U^{-}_{\mathrm{ACCESS}}}{|E|+m_A+k}, \frac{V_{\mathrm{ACCESS}}}{|\mathcal{I}_{\mathrm{den}}|+m_D}\right) - \mathcal{H}\left(\frac{U^{+}_{j}+m_A+k}{|E|+m_A+k}, \frac{V_j+m_D}{|\mathcal{I}_{\mathrm{den}}|+m_D}\right).$$

The ACCESS term decreases with k, while the comparator term increases, so $k = m_D$ gives the most conservative difference. Thus the recovery denominator is 1,538 and the handling denominator is 1,428 at the bound. No assumption about random missingness or the comparator's actual completion is needed. Granting each comparator maximal scores deliberately favors it; the bounds are not newly measured all-planned scores or bootstrap confidence intervals.

Table S22: Worst-case planned-population BSS bounds

Model	ACCESS lower	Direct upper	Serial upper	Policy upper	Min. gap
DeepSeek Flash	65.17	46.58	60.76	41.89	4.41
Qwen2.5-14B	56.88	28.73	38.31	38.45	18.43
Arctic R1-7B	57.40	6.30	6.08	38.39	19.01
Qwen2.5-7B	59.95	30.07	43.09	36.62	16.86
Qwen3.8-27B	63.46	43.69	61.23	40.07	2.23

Note. **Even maximal comparator credit preserves the BSS ordering.** All 2,856 planned requests are considered. The 56 excluded requests earn zero ACCESS credit; comparators receive maximal possible credit. Existing unresolved scores use ACCESS lower and comparator upper endpoints. Scores are percentages; the minimum gap is in percentage points across the three comparators. These are deterministic bounds.

Table S22 gives positive differences for every model and comparator, with a minimum of 2.23 points (Qwen3.8 versus Serial). DeepSeek remains at least 4.41 points above Serial under this completion bound. This result establishes robustness of the BSS ordering to possible scores on the 56 exclusions, conditional on the existing reference definition. Exact all-planned SafeEX, SHR, BSS, strata, and paired intervals require the added request-level references and recorded comparator outcomes; the terminal-cohort values in the main tables remain unchanged.

S17 Constructing Three Specificity Levels

The construction procedure varies the explicitness of a source question before attaching role-conditioned benchmark records. It is an offline evaluation-data procedure, separate from ACCESS's online clarification and pruning. The generator receives the original natural-language question and database schema, including foreign keys and available column descriptions. Gold SQL, benchmark authorization labels, and role permissions are excluded from rewriting inputs. Gold SQL and role policies are subsequently preserved for evaluation.

Source spans define the controllable requirements. The generator identifies nonoverlapping verbatim spans for requested outputs and modifiers such as filters, grouping, ordering, and scope. A separate role check reviews their output/modifier classification. Each span receives a semantic obligation stated in the context of the complete question; shared operators are normalized before editing. An inventory check identifies redundant introductory or population wording already covered by other output spans. Such context remains in the text but is excluded from the counted-fragment budget. Structural checks validate span positions, identities, coverage references, and nonoverlap.

One seeded permutation gives nested selection budgets. For N_{frag} counted fragments, a permutation with seed 42 fixes the order in which specifications are selected for generalization. At target retention $\rho \in \{0, \frac{1}{2}, 1\}$, the retained count is

$$N_{\text{keep}}(\rho, N_{\text{frag}}) = \operatorname*{arg\,min}_{k \in \{0,\ldots,N_{\text{frag}}\}} (|k - \rho N_{\text{frag}}|, k),$$

where k is a candidate retained-fragment count and minimization is lexicographic, preferring the smaller count on a distance tie. Thus L0 retains zero counted specifications in its plan, L1 retains $\lfloor N_{\text{frag}}/2 \rfloor$, and L2 retains all N_{frag}. Let F_ℓ^{hide} be the fragment identifiers selected for generalization at level $\ell \in \{0, 1, 2\}$. These sets satisfy $F_2^{\text{hide}} \subseteq F_1^{\text{hide}} \subseteq F_0^{\text{hide}}$; L2 returns the original question verbatim. These are fragment-selection budgets, not measurements of semantic information or query difficulty. The generalization planner may retain a selected fragment as context when hiding it would damage protected requirements or its specification remains determined elsewhere in the question. The implementation distinguishes the target budget, the rounded planned budget, and the fraction of fragments actually assigned a hiding action.

Generalization preserves a recoverable intended meaning. The default writer returns replacements for selected spans, and the program inserts them at the original positions. Coupled grammatical constructions may require clause-level rewriting; unsuccessful local edits admit a bounded whole-request fallback under the same checks. A hidden specification must become undecidable from the rewrite while the original requirement remains a possible clarification. A protected specification must remain required, rather than merely compatible with the rewrite. Review also checks the population, condition ownership, aggregation, conjunction, negation, and attachment of modifiers. The semantic-review call receives the source question, obligations, rewrite, and hidden/protected identifiers, but not the writer's hiding rationale. Failed or uncertain verdicts do not pass. A bounded repair may revise directions and regenerate the candidate; the recorded generation setting permits one repair.

The same question supports all role conditions. Candidate screening requires complete three-level records, passed generation review, and three distinct action-based effective-retention fractions; selection also uses anchor-retention and clarification-cue heuristics to form a review queue. The final export uses certified entries, not heuristic scores alone. The review contract checks each source obligation as preserved, askable, silently lost, or contradicted, requiring textual evidence and a clarification question for askable obligations. Export joins the certified source question to the original RBAC records by database and normalized original text, retaining the Gold query, role policy, and benchmark label while replacing the request text. The resulting wording is identical across roles at a given level. The independent 120-text sample and its final 117/120 acceptance are reported in Section S9; this sampled acceptance is distinct from automatic contract checks.

The running example makes the three budgets concrete. The BIRD question in the construction example of Section IV-A of the main paper has three counted specifications: post-history count, last-edit date, and the post-title condition. Its exported variants are:

L2: Describe the post history counts and last edit date of the post title "What is the best introductory Bayesian statistics textbook?"

L1: Describe the post history details and a key timestamp of the post title "What is the best introductory Bayesian statistics textbook?"

L0: Describe summary information about the post's activity or history and a key timestamp-related attribute for posts identified by a defined selection criterion.

L1 generalizes the two output specifications and preserves the title condition, giving planned retention 1/3 after rounding. L0 additionally generalizes the title condition; L2 preserves the source text. The underlying database, Gold query, and per-role policy do not change.

Recorded configurations separate generation from evaluation. The three historical generation manifests record model aliases `gpt-5.1` for generation and `gpt-5.5` for review, seed 42, retention targets $[0, 0.5, 1]$, local replacement mode, and one repair. These identify the data-construction calls, not ACCESS's frontend or the five SQL-generation models. The supplied implementation matches the historical Spider and LiveSQLBench implementation fingerprints; the BIRD manifest identifies a different historical implementation. Each dataset's recorded prompts match its own manifest. The cross-dataset entry point adds an exact-source-copy instruction to the frame prompt.

Export checks verify the paired construction invariants. Across the available export pool, each level contains 4,031 role records representing 591 source questions. All L2 texts equal their original reference questions; all selected-fragment sets are nested and obey the L1 rounding rule. Gold, roles, policies, benchmark labels, and other non-level source fields are identical across the three levels, and wording is role-invariant. These pool counts precede the main cohort selection documented in Section S16. The exported effective-retention field is null: the intermediate review export omitted that metadata, although selected identifiers and text remain available. Consequently, planned retention is computed from selected identifiers; an observed effective-retention distribution is not inferred from that field. The supplied generation contracts also pass all 27 existing unit tests with network access disabled. These structural checks complement, rather than replace, the reported human semantic review.